



Eco-Friendly Route Planner

Environmental Studies — Working Prototype & Group Project Guide



Project Idea:

Build a web-based prototype that compares travel options using distance, estimated fuel consumption, and estimated carbon emissions, helping users understand the environmental impact of transportation choices.

1. Project Overview

The recommended prototype is a simple website/web app. A user enters a starting point and destination, and the application compares transport modes such as car, bus, bicycle, and walking. The results can show distance, estimated traveltime, fuel consumption, CO₂ emissions, and an environmental comparison.

2. Division of Work — 3 Group Members

Member	Main Responsibility	Key Tasks	Deliverable
Person 1 Website / UI	Frontend and design	- Homepage, location inputs, buttons - Transport options, result cards - Make the website responsive	Complete and good-looking website interface
Person 2 Logic / Calculations	Make the prototype work	- Distance handling - Travel time estimates - Fuel calculations - CO₂ emissions and comparison logic	Working calculations and comparison system
Person 3 Environment / Documentation	Environmental research, testing and presentation	- Research on emissions & sustainability - Choose emission/fuel assumptions - Test the website, find bugs - Prepare report, PPT and video	Environmental content, testing results and documentation

3. What the Prototype Should Do

Basic user flow: Enter starting location → enter destination → click Find Route → calculate/obtain distance → compare transport modes → calculate fuel and CO₂ → display results → identify the lower-emission option.

Example result layout



Car — Distance: 10 km | Fuel: 0.67 L | CO₂: calculated value



Bus — Distance: 10 km | CO₂: calculated value



Bicycle — Distance: 10 km | Fuel: 0 L | CO₂: 0 during travel: approximately 0



Walking → Distance: 10 km | Fuel: 0 | CO₂: 0 during travel: approximately 0

4. Recommended Technology



HTML — webpage structure



CSS — design, layout, colors and responsive appearance

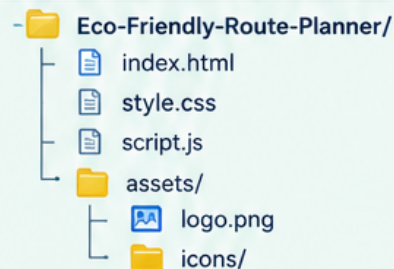


JavaScript — buttons, calculations, comparisons and dynamic results



For a first prototype, you do not need React, Node.js, a database, or a complicated backend.

5. Project Folder Structure



6. Person 1 — Website / Frontend

Create the main webpage with:

- project title
- starting-location box
- destination box
- transport-mode selection
- Find Route button
- route result cards

Suggested homepage text:



EcoRoute

Plan Smarter. Travel Greener.

Compare routes based on distance, fuel consumption and carbon emissions.

Find Route →



Make the design clean, green-themed, readable and mobile-friendly.

6. Person 1 — Website / Frontend

Create the main webpage with: project title; starting-location box; destination box; transport-mode selection; Find Route button; route result cards; environmental-impact section; About Project section.

Suggested homepage text:

■ EcoRoute

Plan Smarter. Travel Greener.

Compare routes based on distance, fuel consumption and carbon emissions.

Make the design clean, green-themed, readable and mobile-friendly.

7. Person 2 — Route and Calculation Logic

This person creates the JavaScript logic. Start with simple predefined/sample routes if a live map service is too difficult. For example, store a few sample locations and their distances. After the basic prototype works, a real routing/map API can be added.

Basic fuel formula:

Fuel used = Distance ÷ Vehicle mileage

Basic emissions formula:

CO₂ = Fuel used × selected emission factor

For buses or other shared transport, use a documented estimated CO₂-per-passenger-kilometre assumption. Keep every assumption clearly stated in the report.

8. Person 3 — Environmental Research and Documentation

Research and explain: transportation-related greenhouse-gas emissions; air pollution; fuel consumption; carbon footprint; sustainable transportation; benefits and limitations of walking, cycling, public transport, carpooling and electric vehicles.

This member should also test the prototype with different routes, record bugs, collect screenshots, prepare the written report, prepare the PPT, and write the demonstration/video script.

9. Recommended Website Sections

Home — Project title, short explanation, and Plan My Route button.

Route Planner — From and To inputs plus Find Routes button.

Comparison — Cards/table comparing transport mode, distance, time, fuel and CO₂.

Environmental Impact — Explain carbon footprint and show estimated CO₂ saved where appropriate.

About the Project — Problem, solution, technology, assumptions, limitations and future scope.

10. Feature That Can Make the Prototype Stand Out

Eco Score: Create a transparent prototype score based on the project's emission assumptions. Example: Car 42/100, Bus 72/100, Bicycle 100/100. Clearly label it as a project-created metric, not an official environmental rating.

CO₂ comparison chart: Show a simple bar chart or horizontal bars for the estimated emissions of each transport mode. This makes the environmental difference easy to understand.

11. Step-by-Step Development Order

1. Create the project folder.
2. Create index.html, style.css and script.js.
3. Person 1 builds the basic HTML interface.
4. Person 1 adds CSS and responsive design.
5. Person 2 adds JavaScript calculations.
6. Connect the Find Route button to the JavaScript logic.
7. Add distance, fuel and CO₂ results.
8. Person 3 adds environmental information and documented assumptions.
9. Add a comparison chart and/or Eco Score.
10. Test multiple routes and transport modes.
11. Take screenshots and record the demonstration.
12. Prepare the report and presentation.

12. Written Project Report Structure

Introduction → Problem Statement → Objectives → Proposed Solution → Methodology → Technologies Used → System Design → Environmental Impact → Results → Limitations → Future Scope → Conclusion → References

13. Presentation / PPT — Suggested 10 Slides

Slide 1: Title — Eco-Friendly Route Planner

Slide 2: Problem — transportation, fuel use and emissions

Slide 3: Why sustainable transportation matters

Slide 4: Our proposed solution

Slide 5: How the Eco-Friendly Route Planner works

Slide 6: Features and user flow

Slide 7: Technology used — HTML, CSS and JavaScript

Slide 8: Prototype screenshots / live demo

Slide 9: Environmental impact, limitations and future scope

Slide 10: Conclusion and team contribution

14. Demonstration Video Plan

Open the website → introduce the project → enter a starting location → enter a destination → click Find Route → show the route/transport comparison → explain distance, fuel and CO₂ → show the lower-emission option → explain why the project can support more environmentally conscious travel choices.

15. Division of Speaking During Presentation

Person 1: Introduction, problem statement and website/interface.

Person 2: Working of the prototype, calculations and technology.

Person 3: Environmental impact, testing, limitations, future scope and conclusion.

16. Possible Future Features

After the basic prototype works, you can add an interactive map, real route distances, live location search, real routing data, weather consideration, EV charging information, carpooling, public-transport data, CO₂ graphs and a more detailed environmental-impact dashboard.

17. Important Project Strategy

Do not start by trying to build Google Maps. First make the minimum working prototype: **Input** → **Distance** → **Transport comparison** → **Fuel** → **CO₂** → **Environmental comparison**. Once this works reliably, add advanced map/API features if your time allows.

18. One-Line Project Definition

“Eco-Friendly Route Planner is a web-based prototype that compares travel options using distance, estimated fuel consumption and carbon emissions to help users understand the environmental impact of transportation choices.”